

LABORATORY PROJECT DOCUMENTATION REPORT

Project: Vehicle Dealership Management System

Course: Database Systems (Spring 2026)

Program: BSCS (Section 5B)

Institution: The University of Lahore, Department of Computer Science & IT



Submitted To: Ms. Ambreen Akmal

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Project Repository: <https://github.com/70152898-DEV/Database-Systems-Term-Project>

1. Project Scenario Description

The **Vehicle Dealership Management System** is a normalized relational database engine designed to streamline operations for local automotive showrooms in Pakistan. Dealerships handle highly valuable, rapidly changing physical assets alongside complex corporate, client, and transactional records.

This project provides a robust backend relational structure to replace error-prone manual logs and unstructured sheets.

1.1 Scope of Operations

The database scope models the real-world lifecycle of vehicles, staff, and transactional processes:

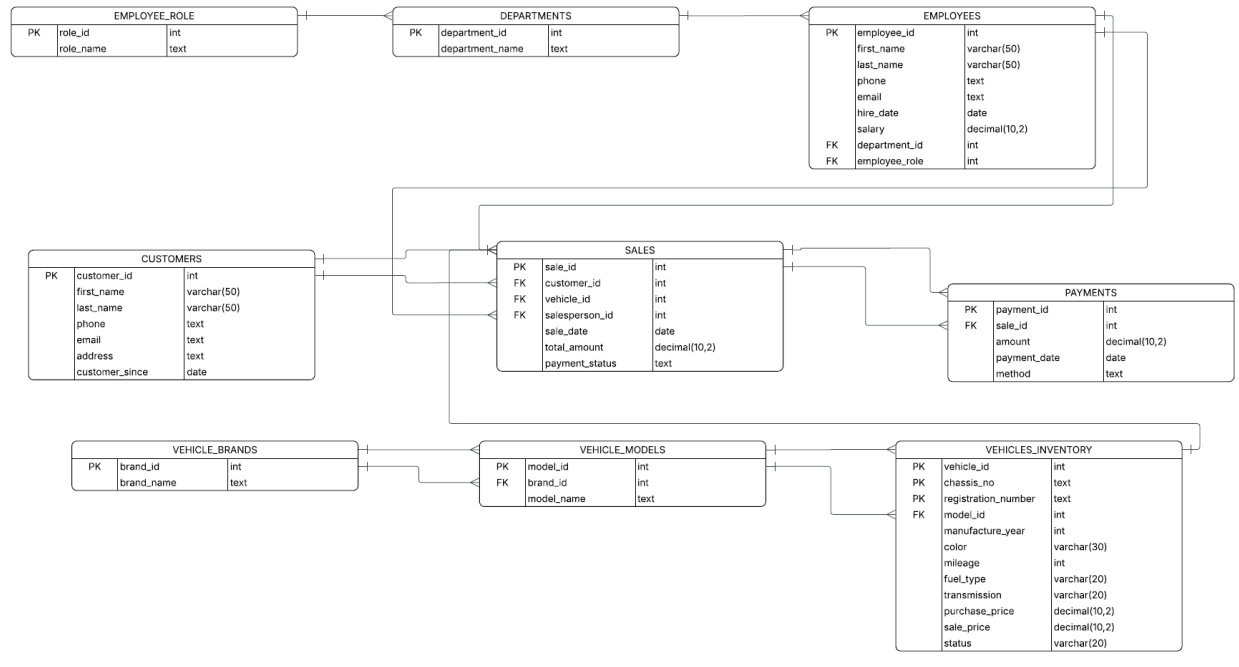
- **The Ingestion Phase:** New vehicles are added to stock with full attributes (chassis number, factory paint color, registration plate code, fuel engine variants, gearbox configurations, showroom dealer purchase pricing, and target showroom floor retail tags).
- **The Staffing Framework:** Employees are onboarded and mapped to explicit departments and roles to maintain organizational balance.
- **The Acquisition Framework:** Showroom customer records are created to hold historical engagement logs and billing address arrays before any transactional agreements are processed.
- **The Sales Contract Lifecycle:** When a vehicle is checked out, the system locks the customer, salesperson, and target vehicle into an unalterable contract table. Because a physical car can only be sold once, a unique structural index constraint is placed on the vehicle identifier to prevent double-selling.
- **The Financial Log Sheet:** Highly valued vehicle transactions are rarely closed via single lump-sum payments. To track financial accounts safely, the system decouples sales contracts from incoming cash flows. This allows accounts managers to log milestone installment payments, bank pay orders, cheques, or cash deposits over time while preserving relational integrity.

2. Database Schema Overview & Architecture

To achieve an optimal relational architecture free of insert, update, and delete anomalies, the system is designed to comply with Third Normal Form (3NF). The schema contains 9 normalized tables:

Lucid Chart: [https://lucid.app/lucidchart/9a002ddd-1e02-4bc5-b51c-](https://lucid.app/lucidchart/9a002ddd-1e02-4bc5-b51c-b72b4d16d0a4/edit?invitationId=inv_021f15ae-6e2b-46f1-bd27-d8445d7da46f)

[b72b4d16d0a4/edit?invitationId=inv_021f15ae-6e2b-46f1-bd27-d8445d7da46f](https://lucid.app/lucidchart/9a002ddd-1e02-4bc5-b51c-b72b4d16d0a4/edit?invitationId=inv_021f15ae-6e2b-46f1-bd27-d8445d7da46f)



2.1 Table Definitions & Key Attributes

- CUSTOMERS:** Holds general profiles of registered clients.
 - Primary Key: customer_id (autoincrementing index sequence).
- DEPARTMENTS:** System lookup directory defining corporate organizational structures.
 - Primary Key: department_id (autoincrementing index sequence).
- EMPLOYEE_ROLE:** System lookup directory defining staffing positions.
 - Primary Key: role_id (autoincrementing index sequence).
- EMPLOYEES:** Showroom staffing directory.
 - Primary Key: employee_id.
 - Foreign Keys: department_id (points to DEPARTMENTS) and employee_role_id (points to EMPLOYEE_ROLE).
- VEHICLE_BRANDS:** Master dictionary for automobile manufacturers (e.g., Toyota, Honda, Suzuki).
 - Primary Key: brand_id.

6. **VEHICLE_MODELS:** Specific vehicle models under master brands.
 - *Primary Key:* model_id.
 - *Foreign Key:* brand_id (points to VEHICLE_BRANDS).
7. **VEHICLES_INVENTORY:** Physical showroom vehicle fleet lot.
 - *Primary Key:* vehicle_id.
 - *Foreign Key:* model_id (points to VEHICLE_MODELS).
8. **SALES:** Central transactional invoice log.
 - *Primary Key:* sale_id.
 - *Foreign Keys:* customer_id (points to CUSTOMERS), salesperson_id (points to EMPLOYEES), and vehicle_id (points to VEHICLES_INVENTORY under a strictly unique constraint).
9. **PAYMENTS:** Real-time ledger tracking individual milestone capitals.
 - *Primary Key:* payment_id.
 - *Foreign Key:* sale_id (points to SALES under a cascading deletion model).

3. Relational Integrity & Mapping Constraints

The database maintains precise relational boundaries:

- **One-to-Many (DEPARTMENTS → EMPLOYEES):** A department can contain many employees, but each employee belongs to exactly one department.
- **One-to-Many (VEHICLE_BRANDS → VEHICLE_MODELS):** A brand can contain many model variants, but a specific model is owned by one brand.
- **One-to-One (VEHICLES_INVENTORY ↔ SALES):** A physical vehicle unit appears on exactly one sales contract. This prevents transactional duplication.
- **One-to-Many (SALES → PAYMENTS):** A sales agreement can accept multiple partial payments, allowing the accounting team to track sequential milestone payments until the balance is cleared.

4. Database Schema DDL Implementation Script

The following script contains the relational creation directives executed within the PostgreSQL engine to build the dealership schema:

```
-- DROP TABLES IN REVERSE ORDER OF DEPENDENCY
DROP TABLE IF EXISTS PAYMENTS CASCADE;
DROP TABLE IF EXISTS SALES CASCADE;
DROP TABLE IF EXISTS VEHICLES_INVENTORY CASCADE;
```

```
DROP TABLE IF EXISTS VEHICLE_MODELS CASCADE;
DROP TABLE IF EXISTS VEHICLE_BRANDS CASCADE;
DROP TABLE IF EXISTS EMPLOYEES CASCADE;
DROP TABLE IF EXISTS EMPLOYEE_ROLE CASCADE;
DROP TABLE IF EXISTS DEPARTMENTS CASCADE;
DROP TABLE IF EXISTS CUSTOMERS CASCADE;
```

```
-- 1. CUSTOMERS TABLE
```

```
CREATE TABLE CUSTOMERS (
    customer_id SERIAL PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    email VARCHAR(100) UNIQUE,
    address TEXT NOT NULL,
    customer_since DATE NOT NULL DEFAULT CURRENT_DATE
);
```

```
-- 2. DEPARTMENTS TABLE
```

```
CREATE TABLE DEPARTMENTS (
    department_id SERIAL PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL UNIQUE
);
```

```
-- 3. EMPLOYEE_ROLE TABLE
```

```
CREATE TABLE EMPLOYEE_ROLE (
    role_id SERIAL PRIMARY KEY,
    role_name VARCHAR(100) NOT NULL UNIQUE
);
```

```
-- 4. EMPLOYEES TABLE
```

```
CREATE TABLE EMPLOYEES (
    employee_id SERIAL PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    hire_date DATE NOT NULL,
    salary NUMERIC(12, 2) NOT NULL CHECK (salary > 0),
    department_id INT NOT NULL,
    employee_role_id INT NOT NULL,
    FOREIGN KEY (department_id) REFERENCES DEPARTMENTS(department_id) ON DELETE
```

```
RESTRICT,  
    FOREIGN KEY (employee_role_id) REFERENCES EMPLOYEE_ROLE(role_id) ON DELETE  
RESTRICT  
);
```

-- 5. VEHICLE BRANDS TABLE

```
CREATE TABLE VEHICLE_BRANDS (  
    brand_id SERIAL PRIMARY KEY,  
    brand_name VARCHAR(100) NOT NULL UNIQUE  
);
```

-- 6. VEHICLE MODELS TABLE

```
CREATE TABLE VEHICLE_MODELS (  
    model_id SERIAL PRIMARY KEY,  
    brand_id INT NOT NULL,  
    model_name VARCHAR(100) NOT NULL,  
    FOREIGN KEY (brand_id) REFERENCES VEHICLE_BRANDS(brand_id) ON DELETE  
CASCADE,  
    CONSTRAINT unique_brand_model UNIQUE (brand_id, model_name)  
);
```

-- 7. VEHICLES INVENTORY TABLE

```
CREATE TABLE VEHICLES_INVENTORY (  
    vehicle_id SERIAL PRIMARY KEY,  
    chassis_no VARCHAR(50) NOT NULL UNIQUE,  
    registration_number VARCHAR(30) UNIQUE,  
    model_id INT NOT NULL,  
    manufacture_year INT NOT NULL,  
    color VARCHAR(30) NOT NULL,  
    mileage INT NOT NULL DEFAULT 0 CHECK (mileage >= 0),  
    fuel_type VARCHAR(30) NOT NULL,  
    transmission VARCHAR(30) NOT NULL,  
    purchase_price NUMERIC(14, 2) NOT NULL CHECK (purchase_price > 0),  
    sale_price NUMERIC(14, 2) NOT NULL CHECK (sale_price > 0),  
    status VARCHAR(30) NOT NULL DEFAULT 'In Stock' CHECK (status IN ('In Stock', 'Sold',  
'Reserved')),  
    FOREIGN KEY (model_id) REFERENCES VEHICLE_MODELS(model_id) ON DELETE  
RESTRICT  
);
```

-- 8. SALES TABLE

```
CREATE TABLE SALES (  

```

```

sale_id SERIAL PRIMARY KEY,
customer_id INT NOT NULL,
vehicle_id INT NOT NULL UNIQUE,
salesperson_id INT NOT NULL,
sale_date DATE NOT NULL DEFAULT CURRENT_DATE,
total_amount NUMERIC(14, 2) NOT NULL CHECK (total_amount > 0),
payment_status VARCHAR(30) NOT NULL DEFAULT 'Pending' CHECK (payment_status IN
('Pending', 'Partial', 'Completed')),
FOREIGN KEY (customer_id) REFERENCES CUSTOMERS(customer_id) ON DELETE
RESTRICT,
FOREIGN KEY (vehicle_id) REFERENCES VEHICLES_INVENTORY(vehicle_id) ON DELETE
RESTRICT,
FOREIGN KEY (salesperson_id) REFERENCES EMPLOYEES(employee_id) ON DELETE
RESTRICT
);

```

-- 9. PAYMENTS TABLE

```

CREATE TABLE PAYMENTS (
    payment_id SERIAL PRIMARY KEY,
    sale_id INT NOT NULL,
    amount NUMERIC(14, 2) NOT NULL CHECK (amount > 0),
    payment_date DATE NOT NULL DEFAULT CURRENT_DATE,
    method VARCHAR(50) NOT NULL CHECK (method IN ('Bank Transfer', 'Pay Order', 'Cash',
'Cheque')),
FOREIGN KEY (sale_id) REFERENCES SALES(sale_id) ON DELETE CASCADE
);

```

```

Query History
93  };
94
95  -- 9. PAYMENTS TABLE
96  CREATE TABLE PAYMENTS (
97      payment_id SERIAL PRIMARY KEY,
98      sale_id INT NOT NULL,
99      amount NUMERIC(14, 2) NOT NULL CHECK (amount > 0),
100     payment_date DATE NOT NULL DEFAULT CURRENT_DATE,
101     method VARCHAR(50) NOT NULL CHECK (method IN ('Bank Transfer', 'Pay Order', 'Cash', 'Cheque')),
102     FOREIGN KEY (sale_id) REFERENCES SALES(sale_id) ON DELETE CASCADE
103  );
104
Data Output  Messages  Notifications
NOTICE: table "payments" does not exist, skipping
NOTICE: table "sales" does not exist, skipping
NOTICE: table "vehicles_inventory" does not exist, skipping
NOTICE: table "vehicle_models" does not exist, skipping
NOTICE: table "vehicle_brands" does not exist, skipping
NOTICE: table "employees" does not exist, skipping
NOTICE: table "employee_role" does not exist, skipping
CREATE TABLE
Query returned successfully in 150 msec.
Total rows:  Query complete 00:00:00.150
CRLF  Ln 104, Col 1

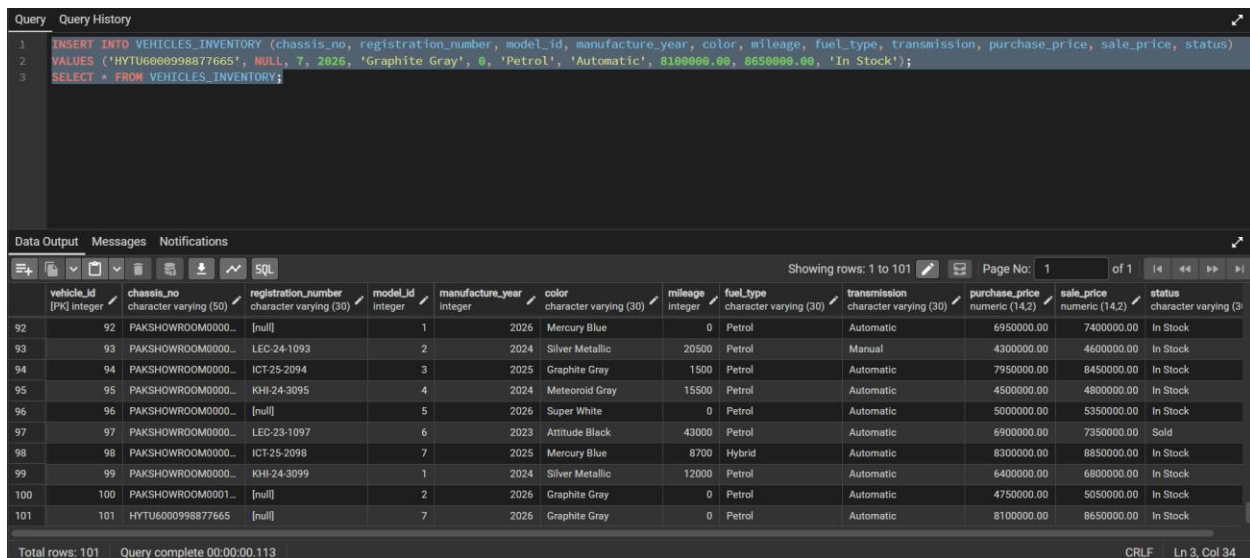
```

5. Implementation of the 10 Project Features & Queries

FEATURE 1: Vehicle Stock Intake

- **Operational Detail:** Inserts a newly acquired car into the showroom fleet table. It maps the vehicle to model ID 7 (Tucson AWD) with detailed specs, setting the status to 'In Stock'.
- **SQL Implementation:**

```
INSERT INTO VEHICLES_INVENTORY (chassis_no, registration_number, model_id,
manufacture_year, color, mileage, fuel_type, transmission, purchase_price, sale_price, status)
VALUES ('HYTU6000998877665', NULL, 7, 2026, 'Graphite Gray', 0, 'Petrol', 'Automatic',
8100000.00, 8650000.00, 'In Stock');
```



The screenshot shows a database query editor with a query history pane on the left and a data output pane on the right. The query history pane shows the following SQL query:

```
1 INSERT INTO VEHICLES_INVENTORY (chassis_no, registration_number, model_id, manufacture_year, color, mileage, fuel_type, transmission, purchase_price, sale_price, status)
2 VALUES ('HYTU6000998877665', NULL, 7, 2026, 'Graphite Gray', 0, 'Petrol', 'Automatic', 8100000.00, 8650000.00, 'In Stock');
3 SELECT * FROM VEHICLES_INVENTORY;
```

The data output pane shows the results of the query in a table. The table has 11 columns: vehicle_id, chassis_no, registration_number, model_id, manufacture_year, color, mileage, fuel_type, transmission, purchase_price, sale_price, and status. The table contains 101 rows of data, with the last row (row 101) corresponding to the vehicle inserted by the query.

vehicle_id	chassis_no	registration_number	model_id	manufacture_year	color	mileage	fuel_type	transmission	purchase_price	sale_price	status
92	PAKSHOWROOM0000...	[null]	1	2026	Mercury Blue	0	Petrol	Automatic	6950000.00	7400000.00	In Stock
93	PAKSHOWROOM0000...	LEC-24-1093	2	2024	Silver Metallic	20500	Petrol	Manual	4300000.00	4600000.00	In Stock
94	PAKSHOWROOM0000...	ICT-25-2094	3	2025	Graphite Gray	1500	Petrol	Automatic	7950000.00	8450000.00	In Stock
95	PAKSHOWROOM0000...	KHI-24-3095	4	2024	Meteoroid Gray	15500	Petrol	Automatic	4500000.00	4800000.00	In Stock
96	PAKSHOWROOM0000...	[null]	5	2026	Super White	0	Petrol	Automatic	5000000.00	5350000.00	In Stock
97	PAKSHOWROOM0000...	LEC-23-1097	6	2023	Attitude Black	43000	Petrol	Automatic	6900000.00	7350000.00	Sold
98	PAKSHOWROOM0000...	ICT-25-2098	7	2025	Mercury Blue	8700	Hybrid	Automatic	8300000.00	8850000.00	In Stock
99	PAKSHOWROOM0000...	KHI-24-3099	1	2024	Silver Metallic	12000	Petrol	Automatic	6400000.00	6800000.00	In Stock
100	PAKSHOWROOM0001...	[null]	2	2026	Graphite Gray	0	Petrol	Automatic	4750000.00	5050000.00	In Stock
101	HYTU6000998877665	[null]	7	2026	Graphite Gray	0	Petrol	Automatic	8100000.00	8650000.00	In Stock

FEATURE 2: Onboard New Personnel

- **Operational Detail:** Registers a new Showroom Consultant inside the sales division (department_id 2, role_id 4) and logs their basic employee details.
- **SQL Implementation:**

```
INSERT INTO EMPLOYEES (first_name, last_name, phone, email, hire_date, salary,
department_id, employee_role_id)
VALUES ('Daniyal', 'Malik', '+923035553344', 'daniyal.malik@dealership.pk', '2026-06-01',
75000.00, 2, 4);
```


Query

Query History

```

1 INSERT INTO EMPLOYEES (first_name, last_name, phone, email, hire_date, salary, department_id, employee_role_id)
2 VALUES ('Daniyal', 'Malik', '+92383553344', 'daniyal.malik@dealership.pk', '2026-06-01', 75000.00, 2, 4);
3
4 SELECT * FROM EMPLOYEES;

```

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 16

Page No: 1 of 1

	employee_id	first_name	last_name	phone	email	hire_date	salary	department_id	employee_role_id
	[PK] Integer	character varying (50)	character varying (50)	character varying (20)	character varying (100)	date	numeric (12,2)	Integer	Integer
6	6	Hamza	Shahzad	+92315666788	h.shahzad@dealership.pk	2023-08-14	110000.00	2	2
7	7	Ayesha	Rehman	+923223334455	a.rehman@dealership.pk	2024-01-10	130000.00	3	3
8	8	Osman	Dawood	+923008889900	o.dawood@dealership.pk	2021-11-05	280000.00	1	1
9	9	Mariam	Javed	+923345556677	m.javed@dealership.pk	2025-05-12	72000.00	2	4
10	10	Faisal	Iqbal	+923451119922	f.iqbal@dealership.pk	2025-09-18	70000.00	2	4
11	11	Khadija	Mansoor	+923218881144	k.mansoor@dealership.pk	2024-07-22	85000.00	2	4
12	12	Umer	Siddiqui	+923014443322	u.siddiqui@dealership.pk	2023-11-30	105000.00	3	3
13	13	Nida	Farooq	+923332228899	n.farooq@dealership.pk	2025-03-15	73000.00	2	4
14	14	Haris	Rauf	+923127774411	h.rauf@dealership.pk	2026-02-10	90000.00	2	2
15	15	Tayyaba	Bano	+923456665544	t.bano@dealership.pk	2024-09-05	78000.00	2	4
16	16	Daniyal	Malik	+923035553344	daniyal.malik@dealership...	2026-06-01	75000.00	2	4

Total rows: 16 Query complete 00:00:00.141

CRLF

Ln 4, Col 25

FEATURE 3: Process Customer Sale

- **Operational Detail:** Connects customer_id 1 with a selected vehicle (vehicle_id 1) and salesperson_id 2 to execute a transaction of 6,600,000.00 PKR.
- **SQL Implementation:**

```
INSERT INTO SALES (customer_id, vehicle_id, salesperson_id, sale_date, total_amount,
payment_status)
VALUES (1, 1, 2, '2026-06-05', 6600000.00, 'Partial');
```

Query Query History

```

1 INSERT INTO SALES (customer_id, vehicle_id, salesperson_id, sale_date, total_amount, payment_status)
2 VALUES (1, 1, 2, '2026-06-05', 6600000.00, 'Partial');
3
4 SELECT * FROM SALES;

```

Data Output Messages Notifications

	sale_id [PK] integer	customer_id integer	vehicle_id integer	salesperson_id integer	sale_date date	total_amount numeric (14,2)	payment_status character varying (30)
1	1	1	1	2	2026-06-05	6600000.00	Partial

Total rows: 1 Query complete 00:00:00.141

CRLF Ln 4, Col 21

FEATURE 4: Log Transaction Payment

- **Operational Detail:** Logs a Bank Transfer of 4,000,000.00 PKR against the customer sale contract created above.
- **SQL Implementation:**

```
INSERT INTO PAYMENTS (sale_id, amount, payment_date, method)
VALUES (1, 4000000.00, '2026-06-05', 'Bank Transfer');
```

Query

Query History

```
1 INSERT INTO PAYMENTS (sale_id, amount, payment_date, method)
2 VALUES (1, 4000000.00, '2026-06-05', 'Bank Transfer');
3
4 SELECT * FROM PAYMENTS;
```

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 1

Page No: 1

of 1

payment_id	sale_id	amount	payment_date	method
[PK] integer	integer	numeric (14,2)	date	character varying (50)
1	1	4000000.00	2026-06-05	Bank Transfer

Total rows: 1

Query complete 00:00:00.147

CRLF

Ln 4, Col 24

FEATURE 5: Update Vehicle Availability Status

- **Operational Detail:** Toggles a vehicle's lot status flag to 'Sold' to reflect its transaction status downstream.
- **SQL Implementation:**

```
UPDATE VEHICLES INVENTORY SET status = 'Sold' WHERE vehicle id = 1;
```

Query

Query History

1

2

3

4

5

UPDATE VEHICLES_INVENTORY

SET status = 'Sold'

WHERE vehicle_id = 1;

SELECT * FROM VEHICLES_INVENTORY;

Data Output

Messages

Notifications

Showing rows: 1 to 101

Page No: 1

of 1

1

4

4

4

4

4

vehicle_id	chassis_no	registration_number	model_id	manufacture_year	color	mileage	fuel_type	transmission	purchase_price	sale_price	status	
[PK] integer	character varying (50)	character varying (30)	integer	integer	character varying (30)	integer	character varying (30)	character varying (30)	numeric (14,2)	numeric (14,2)	character varying (30)	
92	93	PAKSHOWROOM0000...	LEC-24-1093	2	2024	Silver Metallic	20500	Petrol	Manual	4300000.00	4600000.00	In Stock
93	94	PAKSHOWROOM0000...	ICT-25-2094	3	2025	Graphite Gray	1500	Petrol	Automatic	7950000.00	8450000.00	In Stock
94	95	PAKSHOWROOM0000...	KH-24-3095	4	2024	Meteoroid Gray	15500	Petrol	Automatic	4500000.00	4800000.00	In Stock
95	96	PAKSHOWROOM0000...	[null]	5	2026	Super White	0	Petrol	Automatic	5000000.00	5350000.00	In Stock
96	97	PAKSHOWROOM0000...	LEC-23-1097	6	2023	Attitude Black	43000	Petrol	Automatic	6900000.00	7350000.00	Sold
97	98	PAKSHOWROOM0000...	ICT-25-2098	7	2025	Mercury Blue	8700	Hybrid	Automatic	8300000.00	8850000.00	In Stock
98	99	PAKSHOWROOM0000...	KH-24-3099	1	2024	Silver Metallic	12000	Petrol	Automatic	6400000.00	6800000.00	In Stock
99	100	PAKSHOWROOM0001...	[null]	2	2026	Graphite Gray	0	Petrol	Automatic	4750000.00	5050000.00	In Stock
100	101	HYTU600998877665	[null]	7	2026	Graphite Gray	0	Petrol	Automatic	8100000.00	8650000.00	In Stock
101	1	PAKSHOWROOM0000...	LEC-24-1001	1	2024	Super White	15000	Petrol	Automatic	6200000.00	6650000.00	Sold

Total rows: 101

Query complete 00:00:00.140

CRLF

Ln 5, Col 34

FEATURE 6: Adjust Employee Compensation

- **Operational Detail:** Updates an active employee's baseline salary to 85,000.00 PKR following a formal review.
- **SQL Implementation:**

UPDATE EMPLOYEES SET salary = 85000.00 WHERE employee_id = 4;

Query

Query History

1

UPDATE EMPLOYEES

2

SET salary = 85000.00

3

WHERE employee_id = 4;

4

5

SELECT * FROM EMPLOYEES;

Data Output

Messages

Notifications

</

FEATURE 7: Remove Misentered Ledger Records

- **Operational Detail:** Safely deletes a mock error asset row from inventory logs using its unique chassis number.
- **SQL Implementation:**

DELETE FROM VEHICLES_INVENTORY WHERE chassis_no = 'ERR_CHASSIS_99999';

Query

Query History

1

DELETE FROM VEHICLES_INVENTORY

2

WHERE chassis_no = 'ERR_CHASSIS_99999';

3

4

5

SELECT * FROM VEHICLES_INVENTORY;

Data Output

Messages

Notifications

FEATURE 8: Real-Time Active Inventory Search

- **Operational Detail:** Combines the inventory, models, and brands tables via multi-table INNER JOIN logic to render a human-readable list of all 'In Stock' vehicles.
- **SQL Implementation:**

```
SELECT
  v.vehicle_id AS "ID",
  b.brand_name AS "Brand",
  m.model_name AS "Model Spec",
  v.manufacture_year AS "Year",
  COALESCE(v.registration_number, 'Unregistered / Ready for First Owner') AS "Registration State",
  v.color AS "Color",
  v.sale_price AS "Asking Retail (PKR)"
FROM VEHICLES_INVENTORY v
INNER JOIN VEHICLE_MODELS m ON v.model_id = m.model_id
INNER JOIN VEHICLE_BRANDS b ON m.brand_id = b.brand_id
WHERE v.status = 'In Stock';
```

Query Query History

```
1 SELECT
2   v.vehicle_id AS "ID",
3   b.brand_name AS "Brand",
4   m.model_name AS "Model Spec",
5   v.manufacture_year AS "Year",
6   COALESCE(v.registration_number, 'Unregistered / Ready for First Owner') AS "Registration State",
7   v.color AS "Color",
8   v.sale_price AS "Asking Retail (PKR)"
9 FROM VEHICLES_INVENTORY v
10 INNER JOIN VEHICLE_MODELS m ON v.model_id = m.model_id
11 INNER JOIN VEHICLE_BRANDS b ON m.brand_id = b.brand_id
12 WHERE v.status = 'In Stock';
```

Data Output Messages Notifications

Showing rows: 1 to 69

Page No: 1 of 1

ID	Brand	Model Spec	Year	Registration State	Color	Asking Retail (PKR)	
integer	character varying (100)	character varying (100)	integer	character varying	character varying (30)	numeric (14,2)	
59	88	Honda	City Aspire	2026	Unregistered / Ready for First Ow...	Graphite Gray	5100000.00
60	89	Suzuki	Swift GLX	2025	LEC-25-1089	Meteoroid Gray	5050000.00
61	92	Toyota	Corolla Altis X	2026	Unregistered / Ready for First Ow...	Mercury Blue	7400000.00
62	93	Toyota	Yaris ATIV X	2024	LEC-24-1093	Silver Metallic	4600000.00
63	94	Honda	Civic Oriol	2025	ICT-25-2094	Graphite Gray	8450000.00
64	95	Honda	City Aspire	2024	KHI-24-3095	Meteoroid Gray	4800000.00
65	96	Suzuki	Swift GLX	2026	Unregistered / Ready for First Ow...	Super White	5350000.00
66	98	Hyundai	Tucson AWD	2025	ICT-25-2098	Mercury Blue	8850000.00
67	99	Toyota	Corolla Altis X	2024	KHI-24-3099	Silver Metallic	6800000.00
68	100	Toyota	Yaris ATIV X	2026	Unregistered / Ready for First Ow...	Graphite Gray	5050000.00
69	101	Hyundai	Tucson AWD	2026	Unregistered / Ready for First Ow...	Graphite Gray	8650000.00

Total rows: 69 Query complete 00:00:00.126 CRLF Ln 13, Col 1

FEATURE 9: Salesperson Performance Leaderboard

- **Operational Detail:** Uses COUNT, SUM, and GROUP BY aggregates to calculate total vehicle units sold and gross sales revenue managed per staff member, ranking them in descending order.
- **SQL Implementation:**

```

SELECT
    e.employee_id AS "Staff ID",
    CONCAT(e.first_name, ' ', e.last_name) AS "Salesperson Name",
    COUNT(s.sale_id) AS "Units Cleared",
    SUM(s.total_amount) AS "Gross Revenue Handled (PKR)"
FROM EMPLOYEES e
INNER JOIN SALES s ON e.employee_id = s.salesperson_id
GROUP BY e.employee_id, e.first_name, e.last_name
ORDER BY "Gross Revenue Handled (PKR)" DESC;

```

Query		Query History	
1	SELECT		
2	e.employee_id AS "Staff ID",		
3	CONCAT(e.first_name, ' ', e.last_name) AS "Salesperson Name",		
4	COUNT(s.sale_id) AS "Units Cleared",		
5	SUM(s.total_amount) AS "Gross Revenue Handled (PKR)"		
6	FROM EMPLOYEES e		
7	INNER JOIN SALES s ON e.employee_id = s.salesperson_id		
8	GROUP BY e.employee_id, e.first_name, e.last_name		
9	ORDER BY "Gross Revenue Handled (PKR)" DESC;		
10			

Data Output		Messages		Notifications	
Showing rows: 1 to 1		Page No: 1	of 1		
Staff ID	Salesperson Name	Units Cleared	Gross Revenue Handled (PKR)		
Integer	text	bigint	numeric		
1	2 Zainab Fatima	1	6600000.00		

Total rows: 1	Query complete 00:00:00.121	CRLF	Ln 10, Col 1
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FEATURE 10: Comprehensive Customer Purchase History

- **Operational Detail:** A 5-table joint query tracking a client's purchases, total prices, sum of payments made, and outstanding balance remaining to display a clear CRM ledger.
- **SQL Implementation:**

```

SELECT
    CONCAT(c.first_name, ' ', c.last_name) AS "Client Profile",
    c.phone AS "Contact Number",
    s.sale_date AS "Deal Closed On",
    CONCAT(b.brand_name, ' ', m.model_name) AS "Purchased Asset",
    s.total_amount AS "Agreed Contract Price (PKR)",
    COALESCE(SUM(p.amount), 0.00) AS "Total Capital Paid (PKR)",
    (s.total_amount - COALESCE(SUM(p.amount), 0.00)) AS "Outstanding Dues (PKR)",
    s.payment_status AS "Contract Status"
FROM CUSTOMERS c

```

```

INNER JOIN SALES s ON c.customer_id = s.customer_id
INNER JOIN VEHICLES_INVENTORY v ON s.vehicle_id = v.vehicle_id
INNER JOIN VEHICLE_MODELS m ON v.model_id = m.model_id
INNER JOIN VEHICLE_BRANDS b ON m.brand_id = b.brand_id
LEFT JOIN PAYMENTS p ON s.sale_id = p.sale_id
WHERE c.customer_id = 1
GROUP BY c.first_name, c.last_name, c.phone, s.sale_date, b.brand_name, m.model_name,
s.total_amount, s.payment_status;

```

Query

Query History

7

COALESCE(SUM(p.amount), 0.00) AS "Total Capital Paid (PKR)",

8

(s.total_amount - COALESCE(SUM(p.amount), 0.00)) AS "Outstanding Dues (PKR)",

9

s.payment_status AS "Contract Status"

10

FROM CUSTOMERS c

11

INNER JOIN SALES s ON c.customer_id = s.customer_id

12

INNER JOIN VEHICLES_INVENTORY v ON s.vehicle_id = v.vehicle_id

13

INNER JOIN VEHICLE_MODELS m ON v.model_id = m.model_id

14

INNER JOIN VEHICLE_BRANDS b ON m.brand_id = b.brand_id

15

LEFT JOIN PAYMENTS p ON s.sale_id = p.sale_id

16

WHERE c.customer_id = 1

17

GROUP BY c.first_name, c.last_name, c.phone, s.sale_date, b.brand_name, m.model_name, s.total_amount, s.payment_status;

18

Data Output

Messages

Notifications